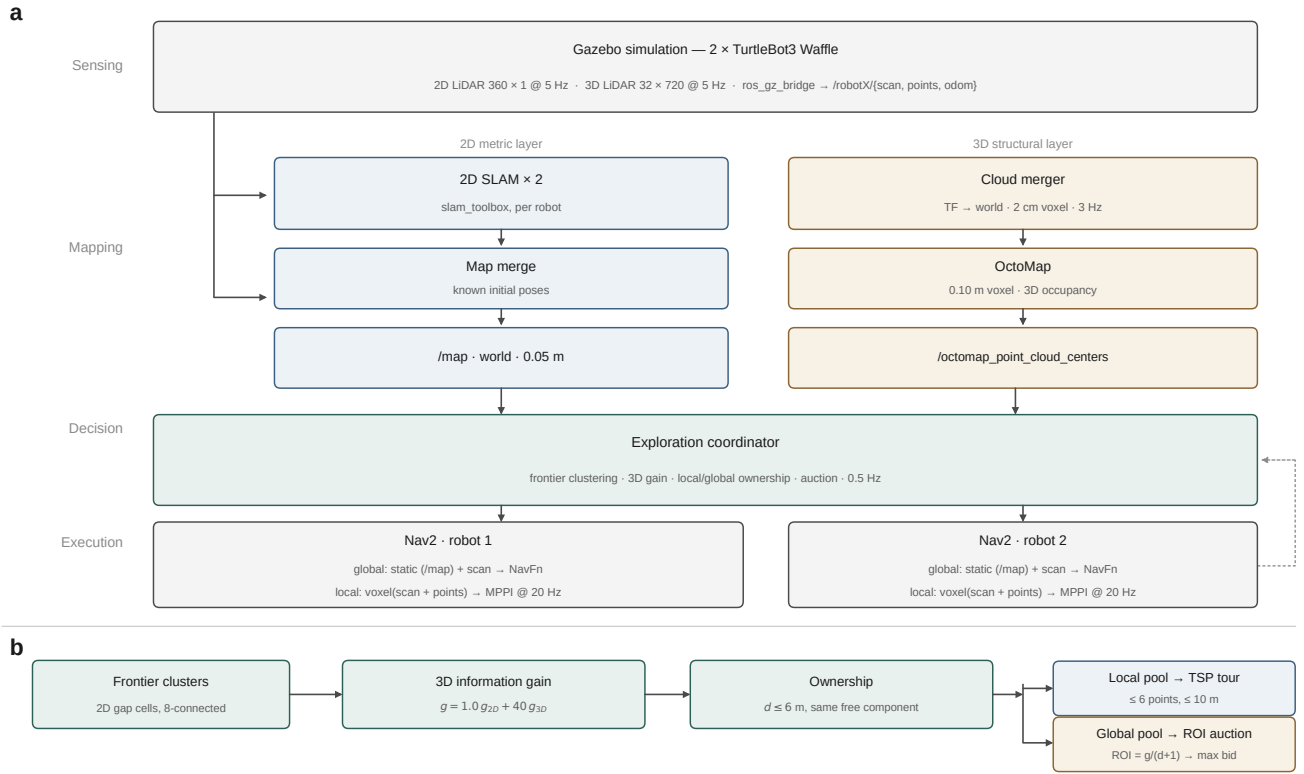
a
Sensing
Gazebo simulation — 2 × TurtleBot3 Waffle
2D LiDAR 360 × 1 @ 5 Hz · 3D LiDAR 32 × 720 @ 5 Hz · ros_gz_bridge → /robotX/{scan, points, odom}
Mapping
2D metric layer
3D structural layer
2D SLAM × 2
slam_toolbox, per robot
Cloud merger
TF → world · 2 cm voxel · 3 Hz
Map merge
known initial poses
OctoMap
0.10 m voxel · 3D occupancy
/map · world · 0.05 m
/octomap_point_cloud_centers
Decision
Exploration coordinator
frontier clustering · 3D gain · local/global ownership · auction · 0.5 Hz
Execution
Nav2 · robot 1
global: static (/map) + scan → NavFn
local: voxel(scan + points) → MPPI @ 20 Hz
Nav2 · robot 2
global: static (/map) + scan → NavFn
local: voxel(scan + points) → MPPI @ 20 Hz
b
Frontier clusters
2D gap cells, 8-connected
3D information gain
$g = 1.0\, g_{2D} + 40\, g_{3D}$
Ownership
$d \le 6$ m, same free component
Local pool → TSP tour
≤ 6 points, ≤ 10 m
Global pool → ROI auction
ROI = g/(d+1) → max bid